EAB 770 Problem Set 1 Key

Miguel Fudolig

1 Problem 1

A batch of 250 pills is tested by randomly sampling 10 pills for quality control. If all 10 pills pass, the batch is approved for distribution. Suppose there are 25 pills that were not acceptable in this batch, what is the probability that the batch passes? Identify the appropriate distribution to calculate these probabilities.

Solution

The most appropriate distribution to calculate these probabilities is the *hypergeometric distribution*. We are interested in P(X=0) when there are $N = 250$ pills, $m = 25$ unacceptable pills, and a sample of $k = 10$ pills.

```
N <- 250
m <- 25
k <- 10

dhyper(x=0,m=m,n=N-m,k=k)
```

```
[1] 0.3415857
```

2 Problem 2

Aseffa et al. (1998) examined the prevalence of HIV among women visiting health care clinics in northwest Ethiopia. Along with testing individuals for HIV, additional information was collected on each woman such as condom use.

Ethiopia HIV and condom use data. Data source is Aseffa et al. (1998).

		HIV		
		Positive	Negative	Total
Condom	Never	135	434	569
	Ever	15	9	24
	Total	150	443	539

2.1 2a

Calculate the following 95% confidence intervals for the difference in probabilities of being HIV positive based on condom use. (Corrected Wald, Agresti-Caffo, Miettinen-Nurminen).

Solution

```
library(DescTools)
```

```
Warning: package 'DescTools' was built under R version 4.6.1
```

```
BinomDiffCI(x1=135,n1=569,
            x2=15,n2=24,
            method=c('waldcc','ac','mn'))
```

```
              est     lwr.ci     upr.ci
waldcc -0.3877417 -0.6062683 -0.1692150
ac     -0.3877417 -0.5674447 -0.1869673
mn     -0.3877417 -0.5556362 -0.1863982
```

2.2 2b

At a significance level of 0.05, do we have any evidence of association between condom use and HIV prevalence?

Solution

There are two possible ways to answer this question: using the confidence intervals in (2a) or using the Mantel-Haenszel test of association.
Because the confidence intervals in (2a) did not include 0, we have strong evidence of an association between condom use and HIV prevalence.
The Mantel-Haenszel tests can be performed as shown:

```r
dat <- expand.grid(Condom=c("Never","Ever"),
                   HIV = c("+","-"))

dat$Count <- c(135,15,434,9)

dat_cc <- xtabs(Count~Condom+HIV, data=dat)

library(vcdExtra)
```

```
Warning: package 'vcdExtra' was built under R version 4.6.1
```

```
Loading required package: vcd
```

```
Warning: package 'vcd' was built under R version 4.6.1
```

```
Loading required package: grid
```

```
Loading required package: gnm
```

```
Warning: package 'gnm' was built under R version 4.6.1
```

```
Registered S3 methods overwritten by 'vcdExtra':
  method      from
  pairs.table vcd
  print.Kappa vcd


Attaching package: 'vcdExtra'
```

```
The following objects are masked from 'package:vcd':

    pairs_diagonal_mosaic, woolf_test
```

```r
CMHtest(dat_cc)
```

```
Cochran-Mantel-Haenszel Statistics for Condom by HIV

                 AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 18.291  1 1.8961e-05
rmeans  Row mean scores differ 18.291  1 1.8961e-05
cmeans  Col mean scores differ 18.291  1 1.8961e-05
general    General association 18.291  1 1.8961e-05
```

The MH test yielded $\chi^2_1 = 18.3, p < 0.0001$. At the 0.05 significance level, we reject the null hypothesis. We have sufficient evidence to support the claim that there is an association between condom use and HIV prevalence.

2.3 2c

Estimate the odds ratio and its corresponding 95% confidence interval. Interpret the odds ratio in the context of the problem.

```
library(epitools)
```

```
Warning: package 'epitools' was built under R version 4.6.1
```

```
Attaching package: 'epitools'
```

```
The following object is masked from 'package:vcdExtra':

    expand.table
```

```
The following object is masked from 'package:vcd':

    oddsratio
```

```
oddsratio(dat_cc,rev="columns")
```

```
$data
       HIV
Condom    -   + Total
  Never 434 135   569
  Ever    9  15    24
  Total 443 150   593

$measure
       odds ratio with 95% C.I.
Condom  estimate    lower    upper
  Never 1.000000       NA       NA
  Ever  5.302773 2.288386 13.02109

$p.value
       two-sided
Condom    midp.exact fisher.exact   chi.square
  Never           NA           NA           NA
  Ever  0.0001059792 9.067182e-05 1.865601e-05
```

```
$correction
[1] FALSE

attr(,"method")
[1] "median-unbiased estimate & mid-p exact CI"
```

Those who reported condom use were estimated to have odds of having HIV that is 5.30 (95% CI: 2.29-13.02) times that of those who did not report condom use.

3 Problem 3

Calculate the sensitivity and the specificity of the test and their 95% confidence intervals.

```
dat <- expand.grid(Test = c("Positive","Negative"),
                   Disorder = c("Positive","Negative"))

dat$Count <- c(123,34,27,189)

dat_cc <- xtabs(Count~Test+Disorder,data=dat)
dat_cc
```

```
          Disorder
Test       Positive Negative
  Positive      123       27
  Negative       34      189
```

Solution

```
library(epiR)
```

```
Warning: package 'epiR' was built under R version 4.6.1
```

```
Loading required package: survival
```

```

Attaching package: 'survival'
```

```
The following object is masked from 'package:epitools':

    ratetable
```

```
Package epiR 2.0.96 is loaded

Type help(epi.about) for summary information

Type browseVignettes(package = 'epiR') to learn how to use epiR for applied epidemiological
```

```
epi.tests(dat_cc)
```

```
          Outcome +    Outcome -      Total
Test +          123           27        150
Test -           34          189        223
Total           157          216        373

Point estimates and 95% CIs:
-------------------------------------------------------------
Apparent prevalence *                  0.40 (0.35, 0.45)
True prevalence *                      0.42 (0.37, 0.47)
Sensitivity *                          0.78 (0.71, 0.85)
Specificity *                          0.88 (0.82, 0.92)
Positive predictive value *            0.82 (0.75, 0.88)
Negative predictive value *            0.85 (0.79, 0.89)
Positive likelihood ratio              6.27 (4.36, 9.00)
Negative likelihood ratio              0.25 (0.18, 0.33)
False T+ proportion for true D- *      0.12 (0.08, 0.18)
False T- proportion for true D+ *      0.22 (0.15, 0.29)
False T+ proportion for T+ *           0.18 (0.12, 0.25)
False T- proportion for T- *           0.15 (0.11, 0.21)
Correctly classified proportion *      0.84 (0.79, 0.87)
-------------------------------------------------------------
* Exact CIs
```

Sensitivity: 0.78 (0.71-0.85)
Specificity: 0.88 (0.82-0.92)

4 Problem 4

In a study by Shaked et al. (2003), researchers studied 26 children with blunt pancreatic injuries. These injuries occurred from a direct blow to the abdomen, bicycle handlebars, fall from height,

or car accident. Nineteen of the patients were classified as having minor injuries, and seven were classified as having major injuries. Pseudocyst formation was suspected when signs of clinical deterioration developed, such as increased abdominal pain, epigastric fullness, fever, and increased pancreatic enzyme levels. In the major injury group, six of the seven children developed pseudocysts while in the minor injury group, three of the 19 children developed pseudocysts.

4.1 4a

Construct the relevant 2x2 table for this problem.

```
dat <- expand.grid(Injury=c("Major","Minor"),
                   Pseudocyst =c("Positive","Negative"))

dat$Count <- c(6,3,1,16)
dat_cc <- xtabs(Count~Injury+Pseudocyst,data=dat)

dat_cc
```

```
       Pseudocyst
Injury  Positive Negative
  Major        6        1
  Minor        3       16
```

4.2 4b

Would it be advisable to use the asymptotic confidence interval when reporting the odds ratio for this problem? Why or why not? If it is not advisable, what confidence interval should we use?

Solution

```
chisq.test(dat_cc)
```

```
Warning in chisq.test(dat_cc): Chi-squared approximation may be incorrect


    Pearson's Chi-squared test with Yates' continuity correction

data:  dat_cc
X-squared = 8.1773, df = 1, p-value = 0.004242
```

Because the chi-squared approximation may be incorrect due to low expected values, it would be better to report the exact confidence intervals.

4.3 4c

Is there sufficient evidence to allow us to conclude that there is an association between pseudocyst development and degree of injuries? Use a significance level of 0.10.

Solution

It is recommended to use the Fisher's Exact test because of the low expected counts.

```
fisher.test(dat_cc)
```

```
    Fisher's Exact Test for Count Data

data:  dat_cc
p-value = 0.002226
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
    2.162745 1539.843484
sample estimates:
odds ratio
  26.16028
```

The resulting p-value is 0.002. We reject the null hypothesis at 0.10 level of significance. There is sufficient evidence to conclude that there is an association between pseudocyst development and degree of injuries.

5 Problem 5

The table below shows data obtained from the 2006 administration of the General Social Survey (GSS).

```
dat <- expand.grid(Work = c("Full","Part"),
                   SelfEmployment = c("No","Yes"),
                   Gender= c("Males","Females"))
```

```r
dat$Count <- c(1051,108,199,38,978,238,87,60)

dat_cc <- xtabs(Count~Work+SelfEmployment+Gender,data=dat)
dat_cc
```

```
, , Gender = Males

      SelfEmployment
Work     No  Yes
  Full 1051  199
  Part  108   38

, , Gender = Females

      SelfEmployment
Work     No  Yes
  Full  978   87
  Part  238   60
```

5.1 5a

At $\alpha = 0.05$, is there evidence of marginal association between work schedule (full-time/part-time) and self-employment?

Solution

Marginal association means we are summing over gender.

```r
dat_marginal <- xtabs(Count~Work+SelfEmployment,data=dat)
CMHtest(dat_marginal)
```

```
Cochran-Mantel-Haenszel Statistics for Work by SelfEmployment

                 AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 29.355  1 6.0276e-08
rmeans  Row mean scores differ 29.355  1 6.0276e-08
cmeans  Col mean scores differ 29.355  1 6.0276e-08
general    General association 29.355  1 6.0276e-08
```

At 0.05 level of significance, we reject the null hypothesis ($\chi^2_1 = 29.4$,p<0.0001). We have sufficient evidence of marginal association between work and self-employment.

5.2 5b

At $\alpha = 0.05$, is there evidence of conditional association between work schedule (full-time/part-time) and self-employment after controlling for gender?

Solution

Conditional association means we have association between work schedule and self-employment at any strata (Gender).

```
CMHtest(dat_cc,overall=T)$ALL
```

```
Cochran-Mantel-Haenszel Statistics for Work by SelfEmployment
    Overall tests, controlling for all strata

                AltHypothesis  Chisq Df      Prob
cor        Nonzero correlation 41.289  1 1.313e-10
rmeans  Row mean scores differ 41.289  1 1.313e-10
cmeans  Col mean scores differ 41.289  1 1.313e-10
general    General association 41.289  1 1.313e-10
```

At 0.05 level of significance, we reject the null hypothesis ($\chi^2_1 = 41.29$,p<0.0001). We have sufficient evidence of conditional association between work and self-employment after controlling for gender.

5.3 5c

Does the assumption of the homogeneity of the odds ratios hold?

Solution

We perform the Breslow-Day test to confirm homogeneity of the odds ratios.

```
breslow_day_test(dat_cc)
```

```
Breslow-Day Test on Homogeneity of Odds Ratios

Data:          dat_cc
OR variables:  Work, SelfEmployment
Strata:        Gender
Common OR:     2.343
```

```
X-squared = 2.3832, df = 1, p-value = 0.1226
```

The resulting test statistic is $\chi^2_1 = 2.38, p = 0.12$. At 0.05 level of significance, we fail to reject the null hypothesis. We have no evidence against the homogeneity of odds ratios. Hence, we can interpret the common odds ratio.

5.4 5d.

Based on your answer in part (5c), report the relevant odds ratios and the asymptotic 95% confidence intervals.

Solution

We can interpret the common odds ratio reported by the `mantelhaen.test()` function.

```
mantelhaen.test(dat_cc,correct=F)
```

```
    Mantel-Haenszel chi-squared test without continuity correction

data:  dat_cc
Mantel-Haenszel X-squared = 41.289, df = 1, p-value = 1.313e-10
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 1.797651 3.053437
sample estimates:
common odds ratio
         2.342864
```

The common odds ratio estimate suggests that the odds of not being self-employed for those who work full time is 2.34 (1.80-3.05) times that of those who work part-time.